MediFlow

Severe Postpartum Hemorrhage (PPH) Prediction System

For Low-Resource Healthcare Settings

Date: March 26, 2026
Dataset: Open-source Maternal Health Dataset (Zimbabwe)

Project Overview

MediFlow is a clinical decision support system designed to predict the likelihood of severe Postpartum Hemorrhage (PPH) in maternity patients, with a specific focus on applicability in low-resource healthcare settings. The system integrates statistical analysis, machine learning, and deep learning into a practical, deployable web application intended to assist frontline healthcare workers in making timely, data-driven decisions.

Core Problem

Postpartum hemorrhage — defined clinically as blood loss exceeding 500 ml after delivery, with severe PPH exceeding 1000 ml — is a leading cause of maternal mortality globally, disproportionately affecting women in low- and middle-income countries where rapid diagnostic tools and specialist support are often unavailable.

Research Questions

1. Which maternal, obstetric, and delivery factors are associated with severe outcomes among women with primary PPH?
2. Can a simple clinical scoring system be developed to predict severe outcomes among patients with primary PPH?
3. Can ML models improve prediction of severe outcomes in primary PPH compared to traditional statistical models?

Core Solution

MediFlow addresses this gap by:

- Predicting severe PPH risk before or at delivery, using structured patient data routinely available in low-resource settings.
- Quantifying individual risk through a clinically interpretable risk score derived from logistic regression odds ratios.
- Estimating blood loss volume (continuous) using a linear regression model, giving clinicians a numerical indicator of hemorrhage severity.
- Deploying the best-performing model as a lightweight web application — a prototype that takes patient data as input and outputs a predicted risk classification with probability score, acting as a supplementary clinical decision support tool.

1. Dataset Description

Field	Detail
Source	Open-source published maternal health dataset — Zimbabwe
Initial Dimensions	223 rows × 42 columns
Final Dimensions	223 rows × 34 columns (after cleaning & encoding)
Target Variable	severe_pph — binary label derived from est_blood_loss_ml > 1000 ml
Class Distribution	158 No Severe PPH (70.9%) / 65 Severe PPH (29.1%)
Key Clinical Features	Parity, booking status, delivery method (NVD, LSCS, Forceps), duration of labour, HIV status

2. Data Cleaning

The raw dataset required extensive preprocessing before modeling.

Step	Action	Details
Missing Values	Imputed with median (numeric) / mode (categorical)	gestational_age_wks: 52 (23.32%); hiv_status_num: 16 (7.17%); duration_labour_min: 9 (4.04%)
Duplicate Rows	Checked and removed	No duplicate rows found
Redundant Columns	Dropped	Columns with identical data (age/age_years, ga/gestational_age_wks, etc.) removed
Column Standardization	Lowercased, underscored	e.g. Est Blood Loss → est_blood_loss_ml
Categorical Encoding	One-hot encoding via pd.get_dummies	booking, delivery_method_clean encoded
Target Engineering	Derived severe_pph column	np.where(est_blood_loss_ml > 1000, 1, 0)

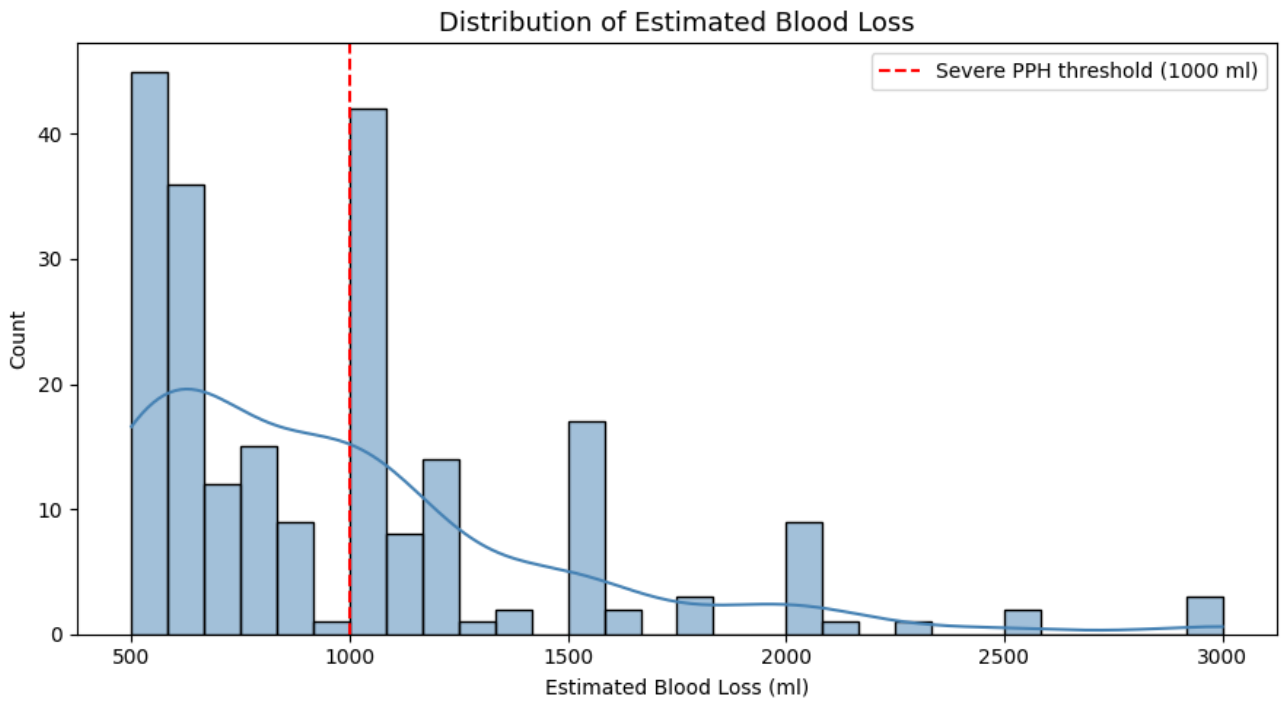
3. Exploratory Data Analysis (EDA)

3.1 Target Variable Distribution

Of 223 patients, 65 (29.1%) were classified as severe PPH (blood loss > 1000 ml). This class imbalance is clinically expected and directly influences model evaluation strategy — Recall (minimizing missed cases) and ROC AUC are prioritized over raw accuracy.

3.2 Blood Loss Distribution

The distribution of est_blood_loss_ml was right-skewed: most patients experienced lower blood loss, with a long tail towards higher volumes corresponding to severe PPH cases.

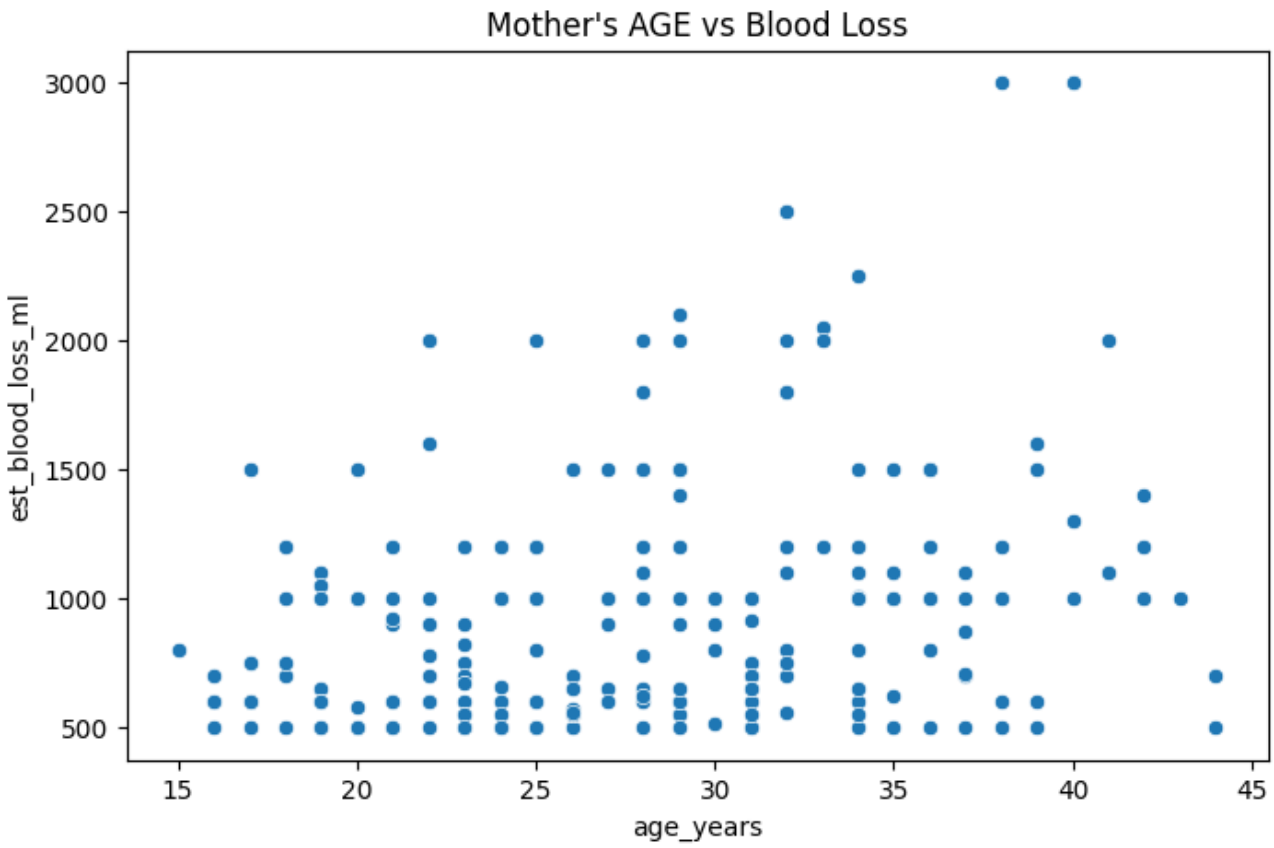


3.3 Scatter Plot Insights

- Labour Duration vs Blood Loss: No strong linear relationship observed; high variance across durations suggests labour duration alone is not a reliable predictor.

- Birth Weight vs Blood Loss: Weak association; birth weight showed minimal direct impact on blood loss.

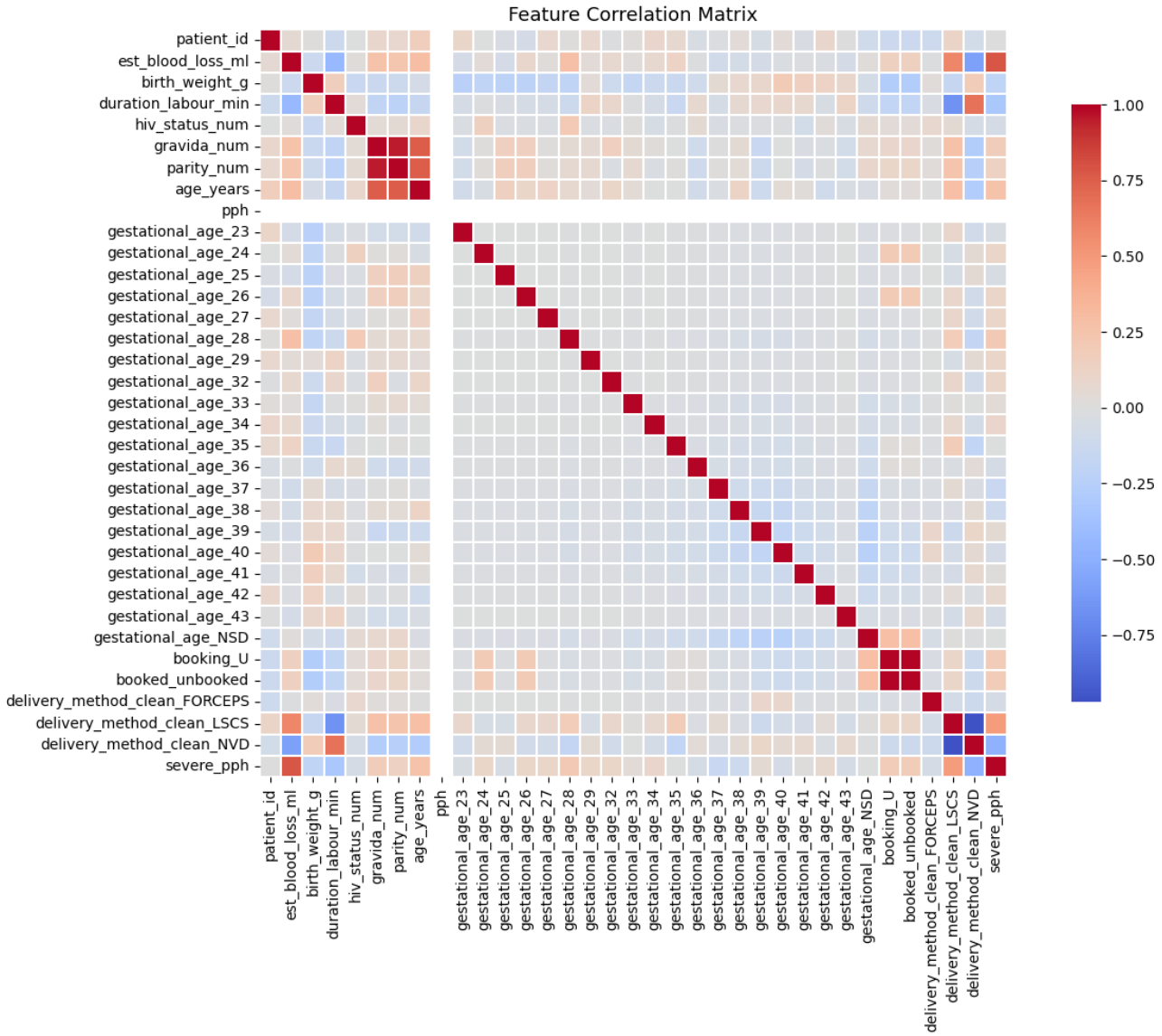
- Maternal Age vs Blood Loss: Some upward trend noted; older mothers showed slightly higher blood loss on average.


Mother's AGE vs Blood Loss
est_blood_loss_ml
3000
2500
2000
1500
1000
500
15
20
25
30
35
40
45
age_years

3.4 Correlation Analysis

A correlation heatmap revealed:

- delivery_method_clean_LSCS was strongly positively correlated with blood loss.
- booked_unbooked showed a moderate association with higher blood loss.
- Several gestational age dummy variables were highly correlated with each other, leading to their removal.



3.5 Multicollinearity Check (VIF)

A Variance Inflation Factor (VIF) analysis confirmed all final features were well below the threshold of 5, indicating no problematic multicollinearity:

Feature	VIF
parity_num	2.15
delivery_method_clean_LSCS	1.78
duration_labour_min	1.42
hiv_status_num	1.24
booked_unbooked	1.13
delivery_method_clean_FORCEPS	1.02

Excluded variables: gravida_num (collinear with parity_num), booking_U (redundant), all gestational age dummies (sparse, high VIF), birth_weight_g, age_years, patient_id.

4. Clinical Risk Score

A logistic regression model was fitted using statsmodels to identify statistically significant predictors and derive a clinically interpretable risk score.

Odds Ratio Table

Predictor	Odds Ratio	p-value	Significant
delivery_method_clean_LSCS	**10.68**	< 0.001	Yes
booked_unbooked	**3.54**	0.018	Yes
hiv_status_num	0.47	0.119	No
duration_labour_min	1.00	0.545	No
parity_num	1.01	0.949	No

Key interpretation:

- LSCS delivery carries 10.7× higher odds of severe PPH compared to NVD — the single strongest predictor.
- Unbooked patients carry 3.5× higher odds — consistent with their higher clinical risk profile.
- HIV status, labour duration, and parity were not statistically significant at $p < 0.05$.

Normalized Risk Score

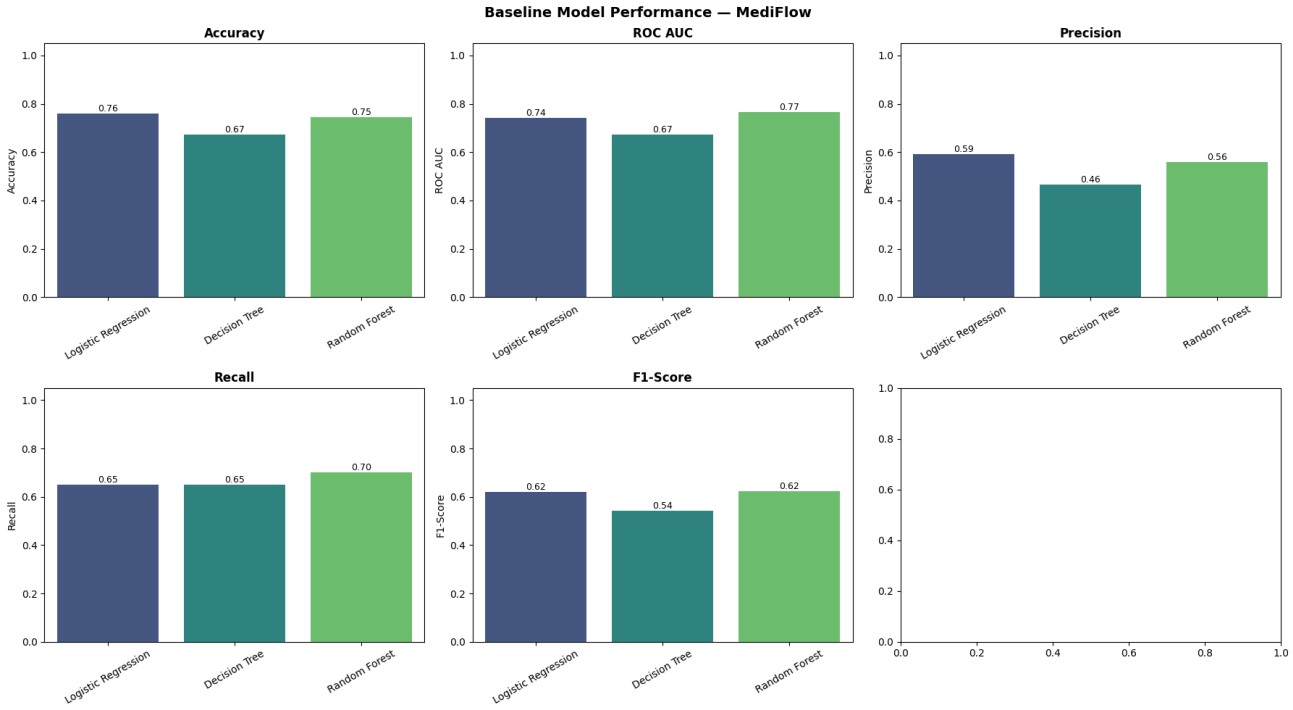
Predictor	Coefficient	Normalized Risk Score
delivery_method_clean_LSCS	2.3684	**1.0000 (reference)**
booked_unbooked	1.2642	0.5338
parity_num	0.0083	0.0035
duration_labour_min	0.0004	0.0001
hiv_status_num	−0.7501	−0.3167

Coefficients are normalized relative to delivery_method_clean_LSCS (score = 1.0), enabling a rapid point-based bedside risk triage using two routinely available clinical data items.

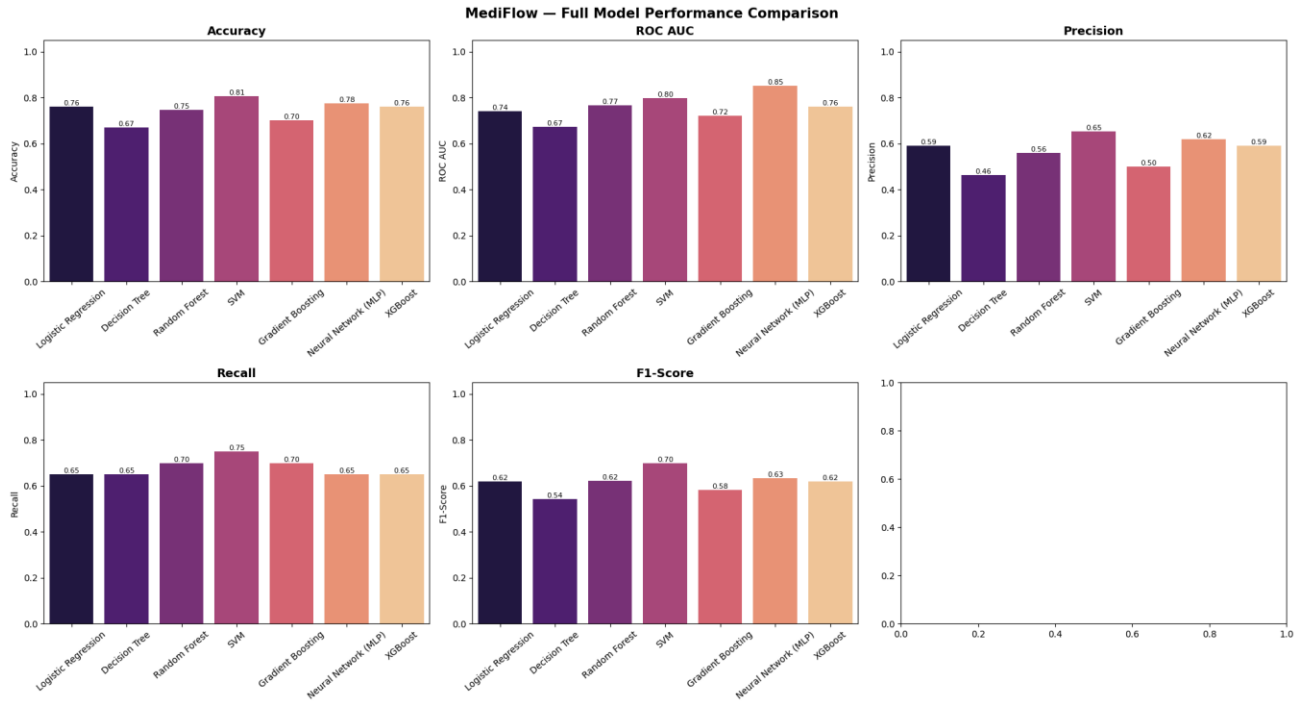
5. Machine Learning & Deep Learning Model Comparison

Seven models were trained and evaluated on a stratified 70/30 train/test split. SVM and MLP used StandardScaler-normalized features. All other models used the raw feature matrix.

Base-Line Model Performance



All Model Performance



Full Results Table

Model	Accuracy	ROC AUC	Precision	Recall	F1-Score
Logistic Regression (baseline)	0.7612	0.7420	0.5909	0.6500	0.6190
Decision Tree	0.6716	0.6734	0.4643	0.6500	0.5417
Random Forest	0.7463	0.7670	0.5600	0.7000	0.6222
SVM	**0.8060**	0.7973	**0.6522**	**0.7500**	**0.6977**
Gradient Boosting	0.7015	0.7202	0.5000	0.7000	0.5833
Neural Network (MLP)	0.7761	**0.8527**	0.6190	0.6500	0.6341
XGBoost	0.7612	0.7601	0.5909	0.6500	0.6190

Best Model Per Metric

Metric	Best Model	Score
Accuracy	SVM	0.8060
ROC AUC	Neural Network (MLP)	0.8527
Precision	SVM	0.6522
Recall	SVM	0.7500

Metric	Best Model	Score
F1-Score	SVM	0.6977

Model Selection: SVM

SVM is selected as the best overall model for the MediFlow prototype, winning on 4 of 5 metrics (Accuracy, Precision, Recall, F1-Score). Its Recall of 0.75 means it correctly identifies 3 in every 4 severe PPH cases — critical in a clinical setting where missed diagnoses carry life-threatening consequences.

The Neural Network (MLP) achieves the highest ROC AUC (0.8527), indicating superior probability calibration and discrimination across all classification thresholds. Both models are strong candidates for deployment.

Architecture Notes

- SVM: RBF kernel, probability=True, trained on StandardScaler-normalized features.
- Neural Network (MLP): 2 hidden layers (64 → 32 neurons), ReLU activation, max 1000 iterations, StandardScaler-normalized features.
- XGBoost: eval_metric='logloss', default tree hyperparameters.

Evaluation Rationale

Given class imbalance (29.1% severe PPH):

- Recall is the primary metric — a false negative (missed severe PPH) is clinically dangerous.
- ROC AUC measures discrimination across all thresholds, robust to class imbalance.
- F1-Score balances precision and recall for the minority class.

6. Blood Loss Prediction (Continuous)

A Linear Regression model was trained to predict the continuous quantity of blood loss (est_blood_loss_ml), complementing the binary classification with a granular severity estimate.

Model Performance

Metric	Value
R^2 (explained variance)	0.2857
MAE (Mean Absolute Error)	258.88 ml

The model explains ~28.6% of variance in blood loss. The MAE of ~259 ml is clinically meaningful: it can broadly differentiate between moderate and severe hemorrhage categories.

Predictor Effects

Variable	Effect on Blood Loss (ml)	Interpretation
delivery_method_clean_LSCS	**+603.05 ml**	Strongest predictor — LSCS substantially increases expected blood loss
age_years	+8.47 ml per year	Older maternal age correlates with slightly higher blood loss
birth_weight_g	−0.04 ml per gram	Marginal inverse effect

Clinical Significance: This model gives frontline staff a quantitative blood loss estimate alongside the binary risk flag — directly supporting decisions around blood product preparation, IV access, and escalation protocols.

7. MediFlow Prototype — Web Application

The best-performing classification model (SVM) will be deployed as a web application providing a Clinical Decision Support Interface.

Planned Architecture

Patient Input Form (Web UI) → Input Validation & Preprocessing → Trained ML Model (serialized with joblib/pickle) → Output: Risk Classification, Probability Score, Clinical Risk Score, Estimated Blood Loss.

Technology Stack

Component	Technology
Backend / Model Serving	Gin (golang) / Python (FastAPI)
Database	PostgreSQL
Model Serialization	joblib
Frontend UI	React (typescript) / Tailwind (css)
Deployment	Docker containers → Render

Input Fields

- Booking status (Booked / Unbooked)

- Delivery method (NVD / LSCS / Forceps)
- Parity (number of previous deliveries)
- Duration of labour (minutes)
- HIV status
- Gestational age

Output

- Risk Classification: Severe PPH Likely / Not Likely
- Probability: e.g. 78.4% probability of severe PPH
- Clinical Risk Score: Normalized 0–1 score
- Estimated Blood Loss: Predicted volume in ml (from linear regression model)

8. Summary of Contributions

Component	Status
Data Cleaning & Preprocessing	Complete
Exploratory Data Analysis	Complete
VIF & Multicollinearity Analysis	Complete
Clinical Risk Score (Logistic Regression)	Complete
Baseline ML Comparison (3 models)	Complete
Extended ML/DL Comparison (7 models total)	Complete
Blood Loss Prediction (Linear Regression)	Complete
Best Model Selection (SVM)	Complete
Web App Prototype (MediFlow)	In Development

9. Next Steps

1. Serialize the SVM model using joblib for deployment.
2. Build the MediFlow web app using Streamlit (recommended for rapid prototyping) or Flask — taking patient inputs and outputting risk classification, probability score, clinical risk score, and estimated blood loss.
3. Validate the prototype with sample patient records from the dataset.
4. Present the working prototype to the lecturer for review.

10. Conclusion

MediFlow goes beyond a simple PPH classification task. It demonstrates a full clinical decision support pipeline:

- A rigorous statistical foundation (VIF analysis, logistic regression, clinically interpreted odds ratios).
- A practical bedside risk scoring tool normalized to LSCS as the reference risk.
- A continuous blood loss estimator (linear regression, MAE = 258.88 ml, $R^2 = 0.286$).
- A comprehensive seven-model comparison, ranging from logistic regression to neural networks.
- SVM identified as the best overall model — Accuracy 0.8060, Recall 0.75, F1-Score 0.6977.
- Neural Network (MLP) achieved the highest ROC AUC (0.8527) — strongest probabilistic discriminator.
- A planned end-to-end web prototype deployable in low-resource clinical settings.

The system is designed with clinical relevance at its core — prioritizing recall and interpretability so that frontline midwives and nurses can act on the output without specialist supervision.

Report prepared for academic review — MediFlow Mini Project.